

TEST 1

ANSWER ALL QUESTIONS

DURATION: 1 HOUR

Write your name, Matrics No. and section on every page of your answer sheets. Scan all your answers and upload the file in .pdf via Blackboard. Make sure you upload the correct file.

QUESTIONS 1 (10 Marks)

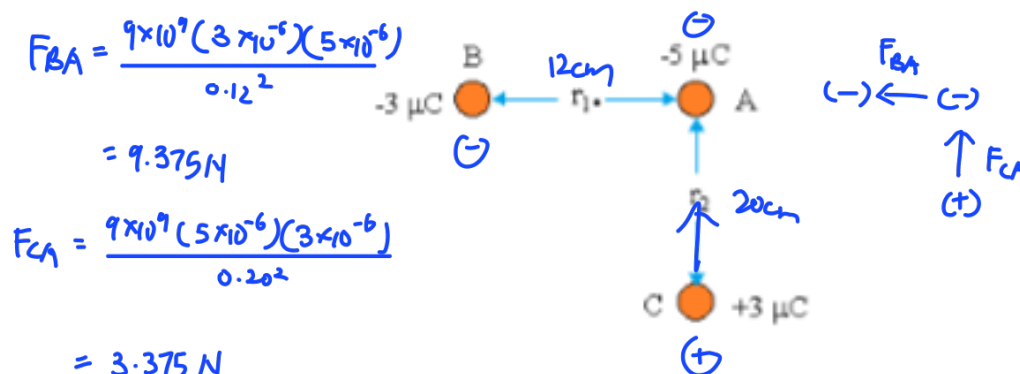
1. a) i. What would happen if a positively charged aluminum rod attracts a suspended globe of straw?

We can conclude that the ball is negatively charged because unlike charges attract each other (1 marks)

- ii. Describe either the electric charge vary or not for the case with mass for the motion of the charged bodies?

No. The charge on an electron is always $1.6 \times 10^{19} \text{C}$ irrespective of its speed. (1 marks)

- b) Based on the Figure 1, calculate the net force on an electric charge A due to other electric charges B and C. Be careful with the direction of the charge, please use Coulomb's Law. The distances are $r_1 = 12.0 \text{ cm}$ and $r_2 = 20.0 \text{ cm}$. $F = 9.95 \text{ N}, 19.8 \text{ degree}$



$$F = \sqrt{3.375^2 + (-9.375)^2} = 9.96 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{3.375}{-9.375} \right) = -19.8^\circ$$

(4 marks)

$$q = \frac{F}{E} = \frac{\frac{mg}{\cos 47^\circ}}{10000} = \frac{\frac{0.02(9.81)}{\cos 47^\circ}}{10000}$$

- c) Figure 2 shows illustrate a globe of mass $m = 0.020 \text{ kg}$ and charge q is tied by a very light string to the roof. The effects of a uniform electric field $E = 10 \text{ kN/C}$ has caused the charged globe to move by an angle of $\theta = 43^\circ$ with the vertical. Draw the free body diagram of the floating globe. Determine the magnitude and the direction of the charge q . -21.03 microC

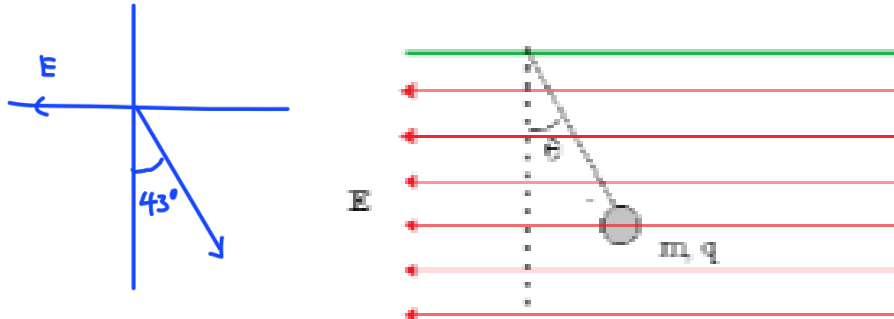


Figure 2

(4 marks)

QUESTIONS 2 (10 Marks)

2. a) Draw diagrams to show the electric field lines due to
i. a positive point charge



(1 marks)

- ii. two equal positive point charges separated by a distance x .



(1 marks)

- b) What are the magnitude and direction of the electric force on an electron in a uniform electric field of strength 1920 N/C that points due east? 3.08 N, west

$$E = \frac{F}{q}, F = Eq = 1920 \times -1.6 \times 10^{-19} = -3.072 \times 10^{-16} \text{ C (to the west)}$$

(2 marks)

- c) An electron acquires $5.25 \times 10^{-16} \text{ J}$ of kinetic energy when it is accelerated by an electric field from plate A to plate B. What is the potential difference between the plates, and which plate is at higher potential? 3281 V, Plate B

$$V = \frac{U}{q} = \frac{5.25 \times 10^{-16}}{-1.6 \times 10^{-19}} = 3281.25 \text{ V}$$

(2 marks)

$U = (+)$, electric potential increase from A to B

Hence, plate B

- d) Figure 3 Two charges $+5.0 \mu\text{C}$ and $-5.0 \mu\text{C}$ are separated by a distance of $2.0 \times 10^{-8} \text{ m}$. The point M is the mid point between the two charges. The point P is $4.0 \times 10^{-8} \text{ m}$ from point M.

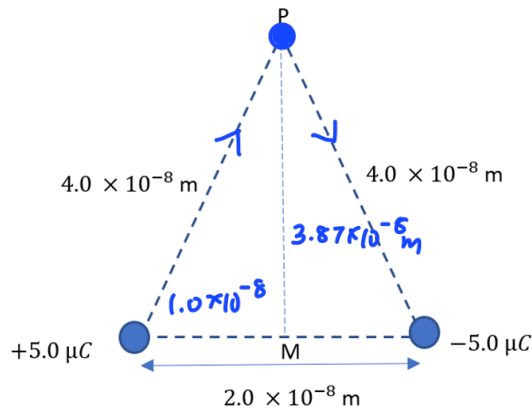
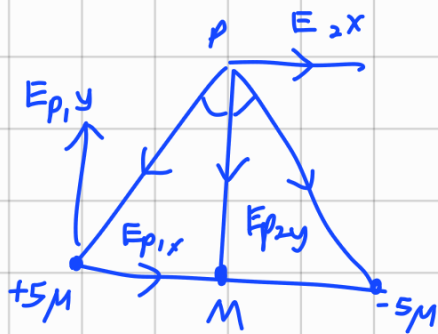


Figure 3

- Calculate the electric field strength at point P. $1.36 \times 10^{19} \text{ N/C}$ (3 marks)
- Calculate the total electric potential at point P. Show your working. 0 V (1 marks)

QUESTIONS 3 (10 Marks)

- Three $6.0 \mu\text{F}$ capacitors are connected in parallel across a 6.0 V battery.
 - What is the equivalent capacitance? 18 microF $C_{\text{eff}} = C_1 + C_2 + C_3 = 6\mu + 6\mu + 6\mu = 18\mu\text{F}$ (1 marks)
 - What is the total charge delivered by the battery? 108 microF $Q = VC = 6(18\mu) = 108\mu\text{C}$ (2 marks)
 - Describe how the capacitance of a parallel-plate capacitor is affected, if any
 - when the potential difference between the plates is increased
When the potential difference between the plates is increased, the capacitance remains the same. (1 marks)
 - when the charge on the capacitor is increased
when the charge on the capacitor is increased, the capacitance remains the same. (1 marks)
 - when the area of each plate is increased
when the charge on the capacitor is increased, the capacitance increases (1 marks)



$$E_{p1} = \frac{9 \times 10^9 (5 \times 10^{-6})}{(4 \times 10^{-8})^2} = E_{p2}$$

$$= 2.8125 \times 10^{19}$$

	X	Y
E_{p1}	$2.8125 \times 10^{19} \cos 76$	$2.6125 \times 10^{19} \sin 73.74$
E_{p2}	$2.8125 \times 10^{19} \cos 76$	$-2.8125 \times 10^{19} \sin 73.74$
	1.36×10^{19}	0

$$\Sigma E = 1.36 \times 10^{19} \text{ N}$$

$$\angle = \cos^{-1}\left(\frac{1}{4}\right)$$

$$= 75.5^\circ$$

$$\approx 76^\circ$$

$$V_A = \frac{kQ}{r} = \frac{k (+5 \times 10^{-6})}{4 \times 10^{-8}}$$

$$V_B = \frac{kQ}{r} = \frac{k (-5 \times 10^{-6})}{4 \times 10^{-8}}$$

$$\Sigma V = V_a + V_b$$

$$= +\frac{kQ}{r} - \frac{kQ}{r}$$

$$= 0V$$

c) A manufacturer wishes to design a capacitor prototype made of $30\text{ }\mu\text{m}$ film of Teflon. If the capacitance is supposed to be $0.5\mu\text{F}$.

i. What should be the effective area of conducting plates?

$$0.81\text{ m}^2$$

$$0.5 \times 10^{-6} = \frac{A(8.85 \times 10^{-12})}{30 \times 10^{-6}}$$

ii. What is the maximum potential difference that the capacitor can withstand?

$$1.8 \times 10^{-3}\text{ V}$$

iii. How much charge can the capacitor hold?

Given that mylar has dielectric constant, $\kappa = 2.1$ and dielectric strength of 60 V/m

$$9.0 \times 10^{-10}\text{ C}$$

(4 marks)

$$Q = VC$$

$$= 1.8 \times 10^{-3} \times 0.5 \times 10^{-6}$$

$$= 9.0 \times 10^{-10}\text{ C}$$